

RETAIL SALES ANALYSIS

1. Project Overview and Objective

This project focuses on analyzing a sales dataset using **Microsoft Excel and Power BI** to derive meaningful business insights. The dataset contains information related to customer orders, product categories, sales amounts, Commission amounts, Discount amounts.

The goal of this project is to transform raw sales data into **interactive dashboards and visual reports** that help in understanding sales performance, identifying trends, and supporting business decision-making.

Using **data modelling, DAX calculations, and visualizations in Power BI**, the project analyses key metrics such as total sales, revenue per order, monthly sales trends, category performance, and geographic sales distribution. These insights help businesses evaluate their performance and identify opportunities for improvement.

Project Objectives

The main objectives of this project are:

Data Preparation

Import and clean the sales dataset using Excel and Power BI.

Data Modeling

Create relationships between tables such as Fact Table and Product Table using the **Product Code** field.

Sales Performance Analysis

Calculate key metrics like:

Total Sales

Monthly Sales

Category wise Sales

Counter wise Sales

Item wise Sales

Total Commission Paid

Commission Paid by Category

Revenue per Order

Year-to-Date (YTD) Sales

Trend Analysis

Analyze **monthly sales trends** to identify growth or decline patterns.

Category Performance Analysis

Evaluate sales performance across different **product categories and sub-categories**.

Interactive Dashboard Creation

Build interactive dashboards using charts such as:

Line charts

Funnel charts

Maps

Matrix tables

Business Insights

Generate meaningful insights that can help organizations improve their sales strategy.

2. Data Sources

Source Description and Timeline:

This is my **Client** Data source is a xlsx dataset containing 32079 records of sales data Between 01/04/2025-13/03/2026.

Domain: Retail Sales Analytics

3. Problem Statement

Many businesses collect large amounts of sales data, but it is often stored in raw formats that are difficult to analyze and interpret. Without proper analysis tools and visualizations, organizations may struggle to identify:

Sales trends over time

High-performing product categories

Underperforming Category and Items

The challenge is to **convert raw sales data into actionable insights** that help organizations improve sales strategies and make data-driven decisions.

This project aims to address this challenge by using **Excel for data preparation and Power BI for data modelling, analysis, and visualization**.

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Date	Date	Unique identifier for each employee
Counter	String (Text)	Name of the Counter
Product Code	String (Text)	Name of the Product Code
HSN Code	Numeric (Integer)	The Product's HSN Code
Product Name	String (Text)	Name of the Product
Category	String (Text)	Product Category
Unit	String (Text)	Product Unit
Qty	Numeric (Integer)	Sold Quantity
MRP	Numeric (Integer)	Maximum Retail Price of the Product
Discount	Numeric (Integer)	The Discount offered on the Product
Rate	Numeric (Integer)	Department where employee works
Commission	Numeric (Integer)	The Commission amount paid for each Item

5. Tools & Technologies

Excel: Data cleaning, transformation, and Pivot Tables.

Power BI: Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

Data Cleaning & Transformation: Removed duplicates, handled missing values, standardized formats, and created calculated fields.

Using Power Query

Date - Change to Date

Counter - Transformation – UPPERCASE

Product Code - =LEFT (B2,4)

Product Name - Trim, Clean (Found some products are with space)

Category - Replace Values (Because total no of category is 30, the category is unique differing only in its tax classification, so product name is alone is sufficient, so we can proceed to replace the data accordingly)

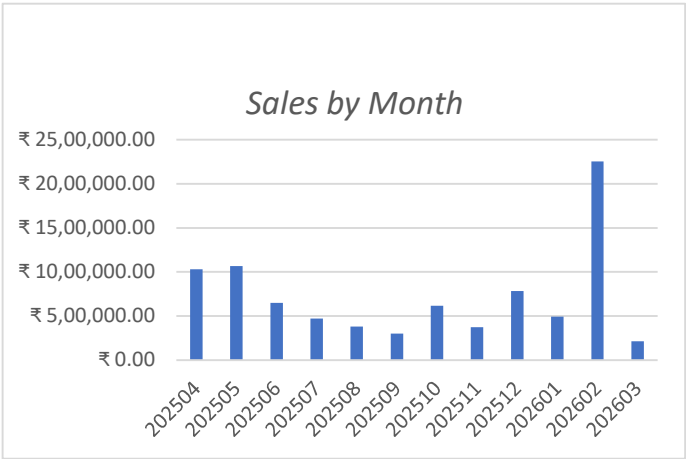
Amount - Qty * Rate

Month - =TEXT(A2,"mmmm yyyy")

Filtering & Sorting: Organized data to focus on relevant records.
Pivot Tables: Generated Pivot Tables for data summarization and initial insights.

Sales by Month

Sales by Month	
Month	Sales Amount
202504	₹ 10,32,503.00
202505	₹ 10,65,520.00
202506	₹ 6,50,630.00
202507	₹ 4,73,416.00
202508	₹ 3,79,767.00
202509	₹ 3,02,834.00
202510	₹ 6,17,141.00
202511	₹ 3,72,614.00
202512	₹ 7,82,855.00
202601	₹ 4,91,484.00
202602	₹ 22,51,892.00
202603	₹ 2,11,768.00
Grand Total	₹ 86,32,424.00



Analysis by Month	
Mean	7,19,368.67
Standard Error	1,59,660.14
Median	5,54,312.50
Mode	#N/A
Standard Deviation	5,53,078.94
Sample Variance	#####
Kurtosis	5.52
Skewness	2.17
Range	20,40,124.00
Minimum	2,11,768.00
Maximum	22,51,892.00
Sum	86,32,424.00
Count	12.00
Confidence Level(95.0%)	3,51,409.59

Month
April 2025
August 2025
December 2025
February 2026
January 2026
July 2025
June 2025
March 2026
May 2025
November 2025
October 2025
September 2025

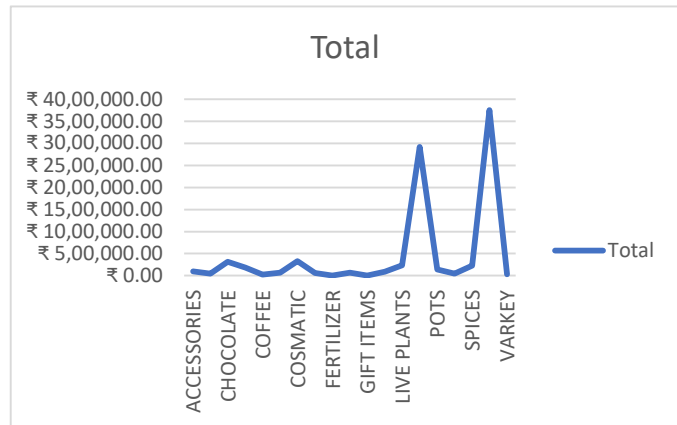
Sales by Category

Sales by Category

Category	Sales Amount
ACCESSORIES	₹ 97,933.00
APPITIZER	₹ 45,360.00
CHOCOLATE	₹ 3,20,828.00
CLOTH	₹ 1,78,339.00
COFFEE	₹ 24,768.00
COOKIES	₹ 62,565.00
COSMATIC	₹ 3,32,928.00
FANCY	₹ 58,590.00
FERTILIZER	₹ 4,199.00
GAME	₹ 63,190.00
GIFT ITEMS	₹ 135.00
HERBALS	₹ 85,274.00
LIVE PLANTS	₹ 2,36,443.00
NATURAL OIL	₹ 29,20,275.00
POTS	₹ 1,39,384.00
POTS WITH PLANT	₹ 43,678.00
SPICES	₹ 2,29,416.00
TEA	₹ 37,59,202.00
VARKEY	₹ 29,917.00
Grand Total	₹ 86,32,424.00

Analysis by Category

Mean	4,54,338.11
Standard Error	2,36,598.72
Median	85,274.00
Mode	#N/A
Standard Deviation	10,31,309.90
Sample Variance	#####
Kurtosis	7.17
Skewness	2.84
Range	37,59,067.00
Minimum	135.00
Maximum	37,59,202.00
Sum	86,32,424.00
Count	19.00
Confidence Level(95.0%)	4,97,075.46

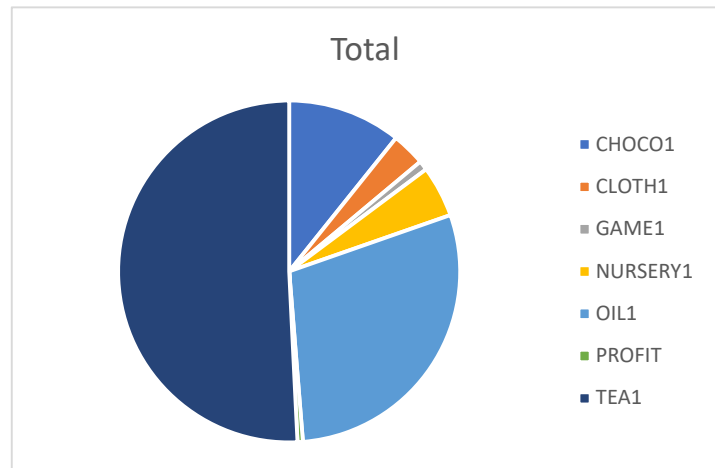


Category
ACCESSORIES
APPITIZER
CHOCOLATE
CLOTH
COFFEE
COOKIES
COSMATIC
FANCY
FERTILIZER
GAME
GIFT ITEMS
HERBALS
LIVE PLANTS
NATURAL OIL
POTS

Sales by Counter

Sales by Counter

Counter	Sales Amount
CHOCO1	₹ 9,28,146.00
CLOTH1	₹ 2,72,490.00
GAME1	₹ 77,010.00
NURSERY1	₹ 4,18,771.00
OIL1	₹ 25,07,921.00
PROFIT	₹ 45,913.00
TEA1	₹ 43,82,173.00
Grand Total	₹ 86,32,424.00



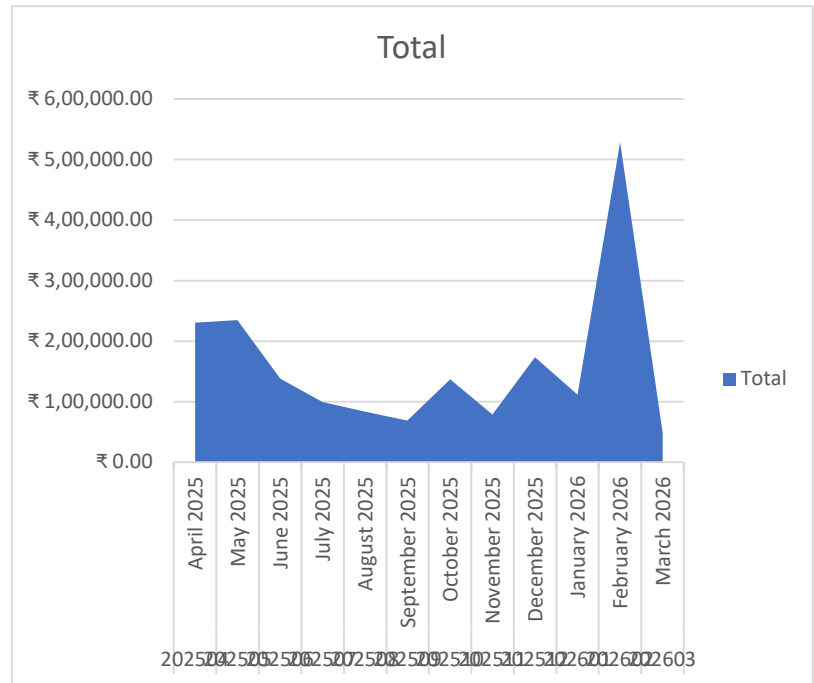
Analysis by Month

Mean	12,33,203.43
Standard Error	6,16,570.52
Median	4,18,771.00
Mode	#N/A
Standard Deviation	16,31,292.26
Sample Variance	#####
Kurtosis	1.59
Skewness	1.54
Range	43,36,260.00
Minimum	45,913.00
Maximum	43,82,173.00
Sum	86,32,424.00
Count	7.00
Confidence Level(95.0%)	15,08,693.71

Counter
CHOCO1
CLOTH1
GAME1
NURSERY1
OIL1
PROFIT
TEA1

Commission by Month

Month	Commission Paid
202504	₹ 2,30,602.00
April 2025	₹ 2,30,602.00
202505	₹ 2,34,846.00
May 2025	₹ 2,34,846.00
202506	₹ 1,37,687.00
June 2025	₹ 1,37,687.00
202507	₹ 99,661.00
July 2025	₹ 99,661.00
202508	₹ 83,337.00
August 2025	₹ 83,337.00
202509	₹ 68,551.00
September 2025	₹ 68,551.00
202510	₹ 1,36,647.00
October 2025	₹ 1,36,647.00
202511	₹ 78,713.00
November 2025	₹ 78,713.00
202512	₹ 1,73,343.00
December 2025	₹ 1,73,343.00
202601	₹ 1,11,254.00
January 2026	₹ 1,11,254.00
202602	₹ 5,28,825.00
February 2026	₹ 5,28,825.00
202603	₹ 48,547.00
March 2026	₹ 48,547.00
Grand Total	₹ 19,32,013.00



Commission Analysis by Month

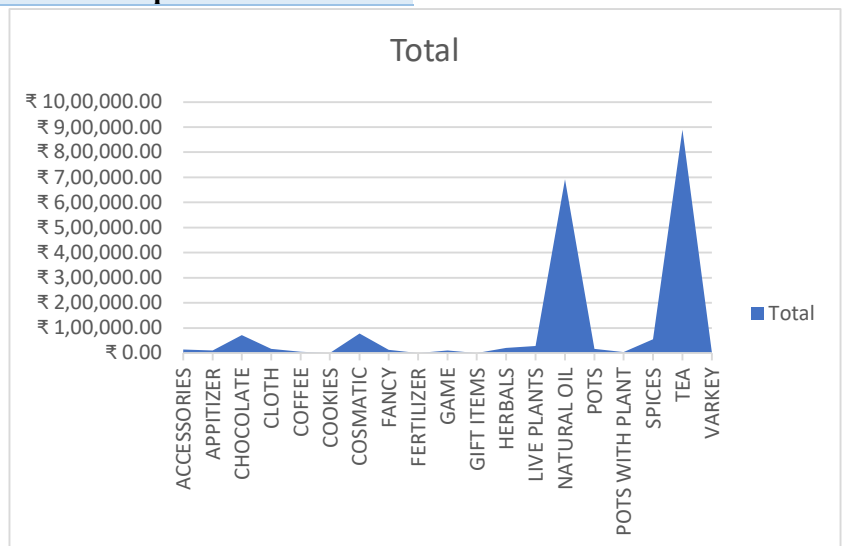
Mean	1,61,001.08
Standard Error	26,037.47
Median	1,23,950.50
Mode	2,30,602.00
Standard Deviation	1,27,557.03
Sample Variance	#####
Kurtosis	4.54
Skewness	2.14
Range	4,80,278.00
Minimum	48,547.00
Maximum	5,28,825.00
Sum	38,64,026.00
Count	24.00
Confidence Level(95.0%)	53,862.61

Month
April 2025
August 2025
December 2025
February 2026
January 2026
July 2025
June 2025
March 2026
May 2025
November 2025
October 2025
September 2025

Commission by Category

Category	Commission paid
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ACCESSORIES	₹ 14,013.00
APPITIZER	₹ 10,383.00
CHOCOLATE	₹ 71,364.00
CLOTH	₹ 16,675.00
COFFEE	₹ 5,451.00
COOKIES	₹ 0.00
COSMATIC	₹ 77,328.00
FANCY	₹ 13,208.00
FERTILIZER	₹ 491.00
GAME	₹ 10,740.00
GIFT ITEMS	₹ 30.00
HERBALS	₹ 19,919.00
LIVE PLANTS	₹ 28,309.00
NATURAL OIL	₹ 6,93,083.00
POTS	₹ 16,750.00
POTS WITH PLANT	₹ 5,246.00
SPICES	₹ 54,585.00
TEA	₹ 8,90,911.00
VARKEY	₹ 3,527.00
Grand Total	₹ 19,32,013.00



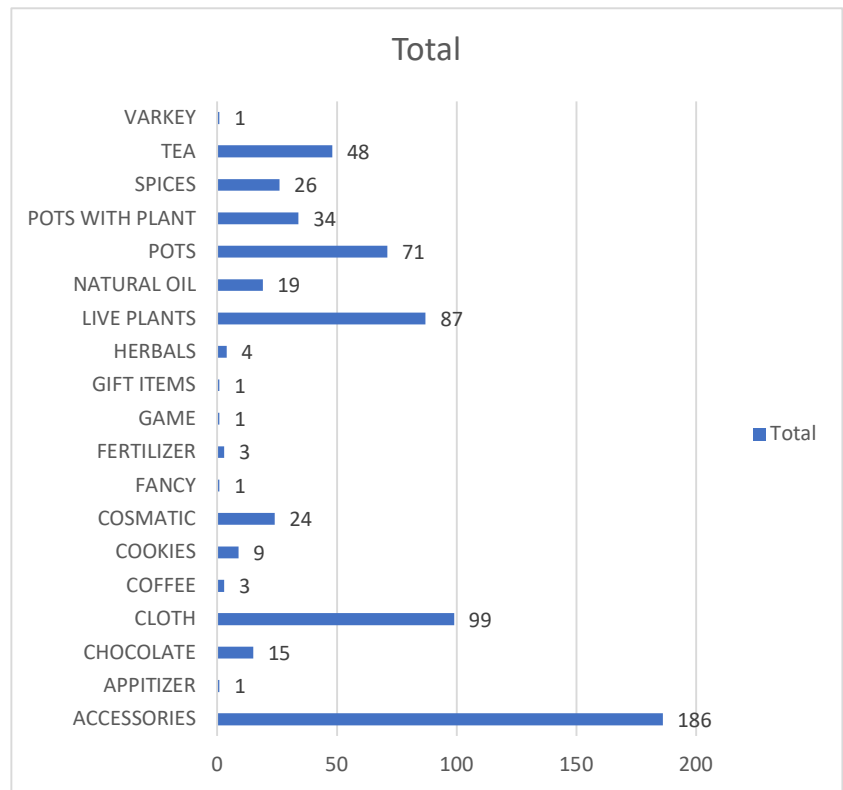
Category
ACCESSORIES
APPITIZER
CHOCOLATE
CLOTH
COFFEE
COOKIES
COSMATIC
FANCY
FERTILIZER
GAME
GIFT ITEMS
HERBALS
LIVE PLANTS
NATURAL OIL
POTS
POTS WITH PLANT
SPICES
TEA
VARKEY

Commission Analysis by category

Mean	1,01,684.89
Standard Error	56,560.85
Median	14,013.00
Mode	#N/A
Standard Deviation	2,46,543.05
Sample Variance	#####
Kurtosis	7.17
Skewness	2.85
Range	8,90,911.00
Minimum	-
Maximum	8,90,911.00
Sum	19,32,013.00
Count	19.00
Confidence Level(95.0%)	1,18,829.95

No of Items by Category

Category	No of Items
ACCESSORIES	186
APPITIZER	1
CHOCOLATE	15
CLOTH	99
COFFEE	3
COOKIES	9
COSMATIC	24
FANCY	1
FERTILIZER	3
GAME	1
GIFT ITEMS	1
HERBALS	4
LIVE PLANTS	87
NATURAL OIL	19
POTS	71
POTS WITH PLANT	34
SPICES	26
TEA	48
VARKEY	1
Grand Total	633



Convert the data into Fact and Dimension Table (If Possible)

1 Fact Table

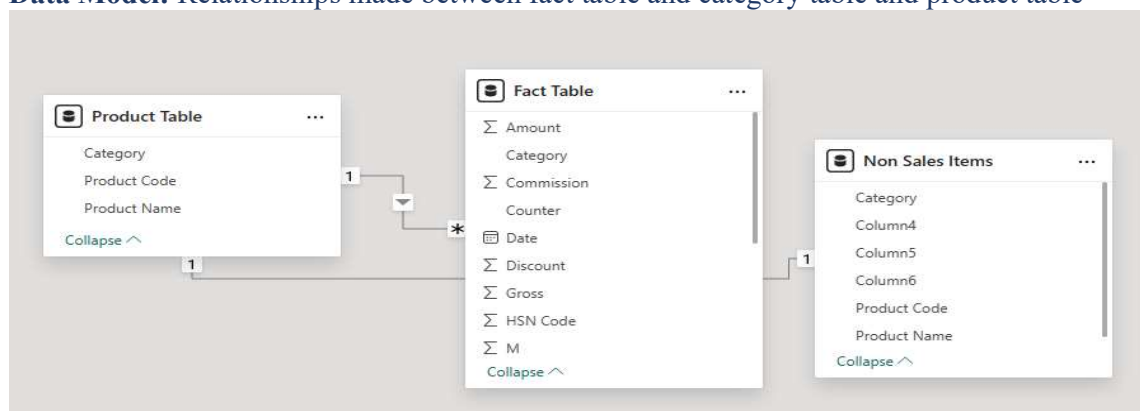
2. Dimension Tables

a. category Table

b. Product Table

7. Data Modelling and DAX (Power BI)

Data Model: Relationships made between fact table and category table and product table



Calculated Columns & DAX Measures: Implemented DAX formulas for key metrics, such as total sales, growth percentages, and cumulative totals

Total Sales with Units

Measure = SUM('Fact Table'[Qty])

Category	Sum of Amount	Unit
ACCESSORIES	97933	761
APPITIZER	45360	63
CHOCOLATE	320828	222867
CLOTH	178339	214
COFFEE	24768	166
COOKIES	62565	89666
COSMATIC	332928	1538
FANCY	58590	434
FERTILIZER	4199	35
GAME	63190	316
GIFT ITEMS	135	1
HERBALS	85274	328
LIVE PLANTS	236443	1740
NATURAL OIL	2920275	11705
POTS	139384	975
POTS WITH PLANT	43678	170
SPICES	229416	1464
TEA	3759202	17653
VARKEY	29917	272

Total Sales

Total Sales = sum('Fact Table'[Amount])

Avg_Sales

Avg_sales=AVERAGE('Fact Table'[Amount])

Year-to-Date (YTD) Sales

YTD_Sales = TOTALYTD('Fact Table'[Total Sales],'Fact Table'[Date])

Month-to-Date (MTD) Sales

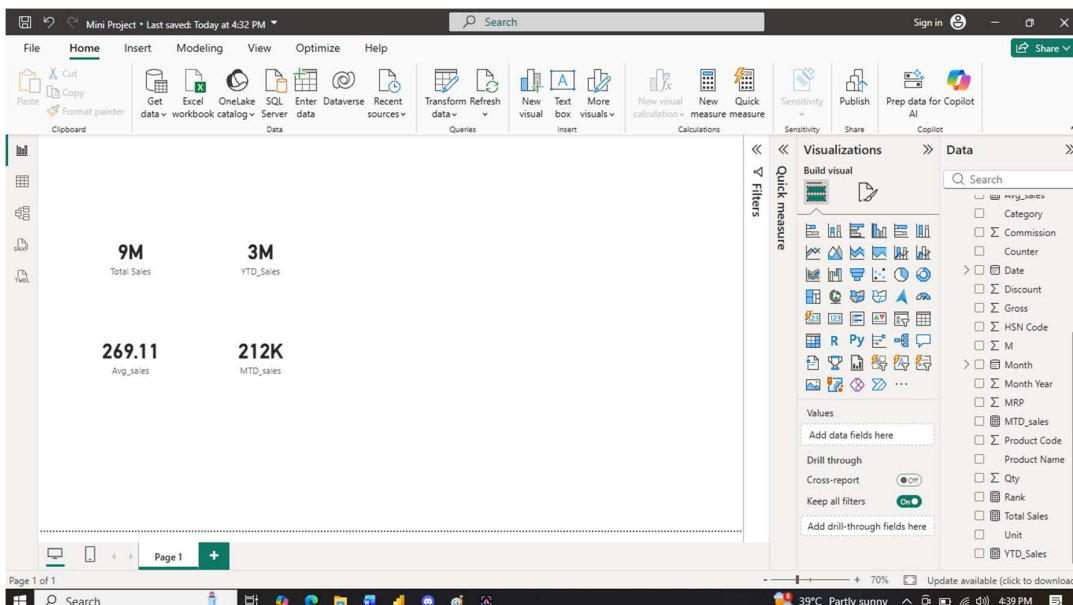
MTD_sales = TOTALMTD([Total Sales],'Fact Table'[Date])

Is there any item that has not been sold this year?

Yes – Total : 465 Products

Using (PowerBi)

IF(COUNTIF(Table_Fact_1!H2:H32079,B2)=0,"Missing in Table", "Available")

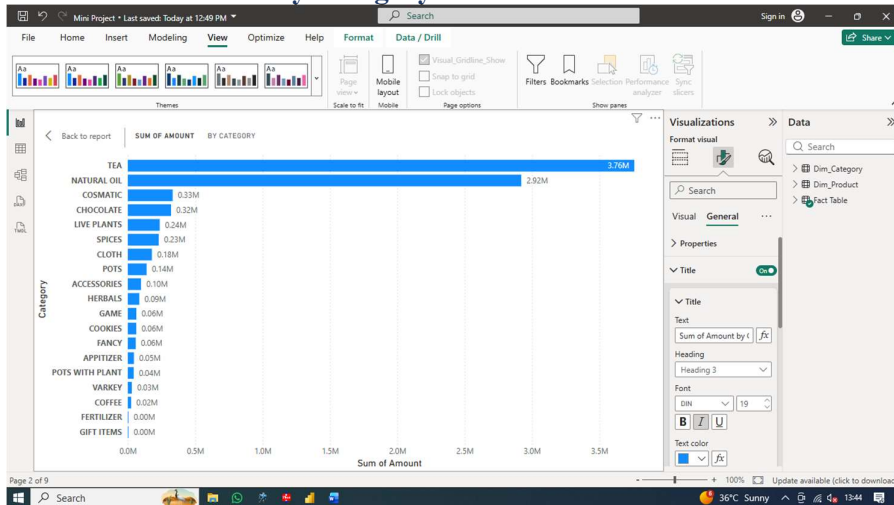


8. Analysis and Visualizations (Power BI)

Dashboard Features:

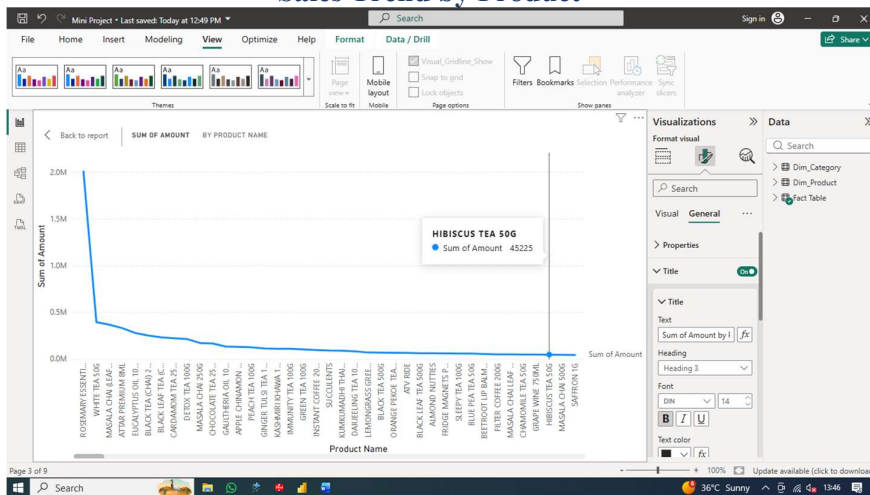
Bar Charts

Sales by Category



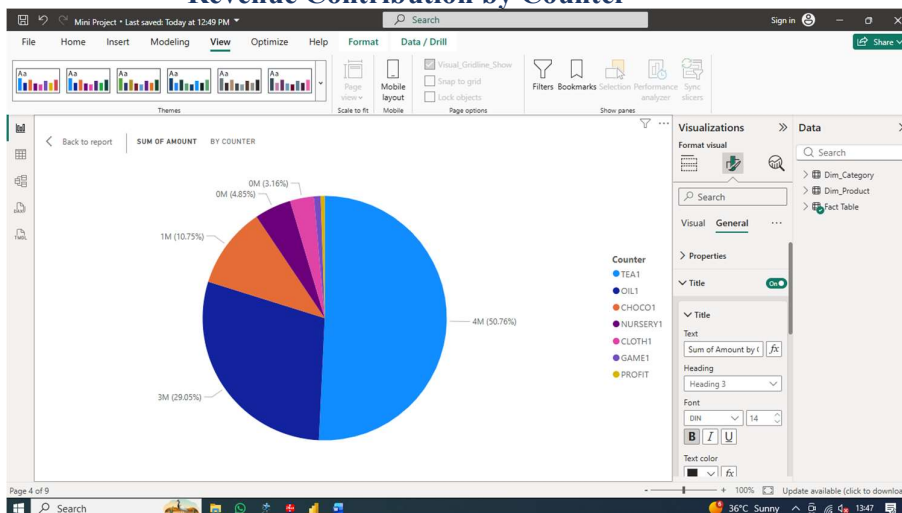
Line Chart

Sales Trend by Product



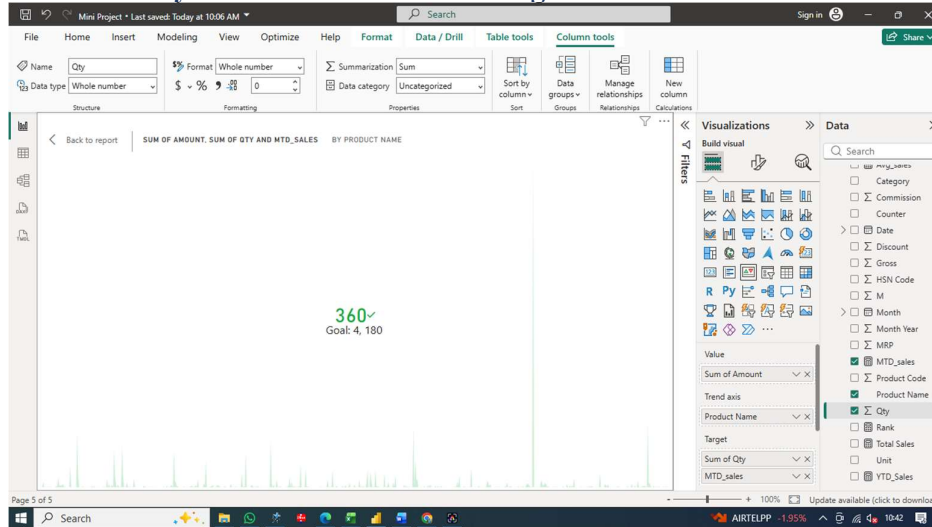
Pie Chart

Revenue Contribution by Counter

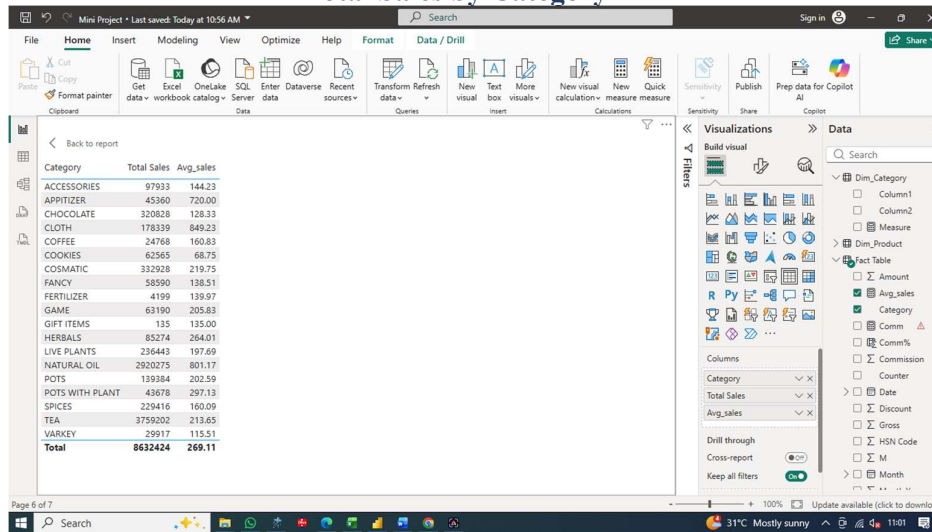


KPI Cards / Summary Metrics

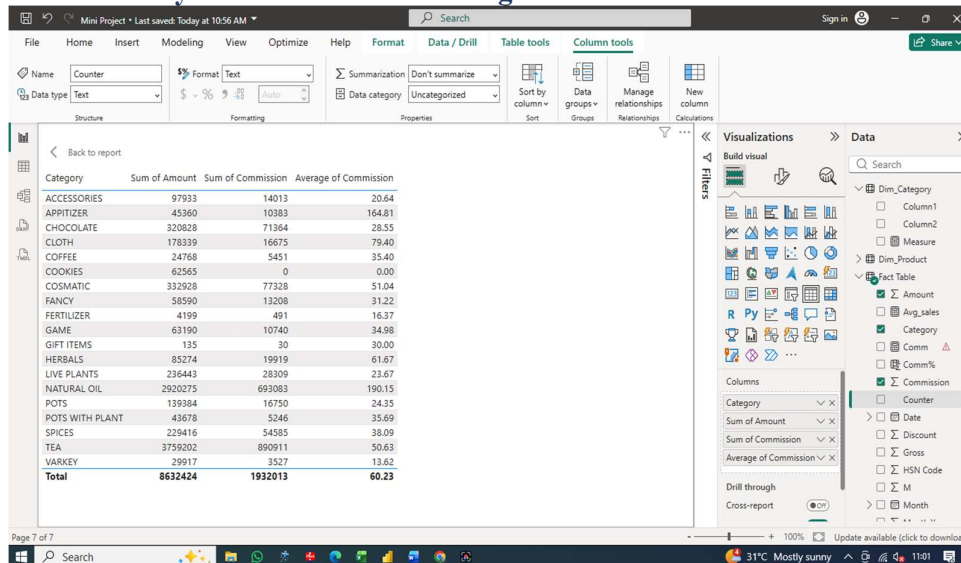
Total Sales by No of Products and Average sales



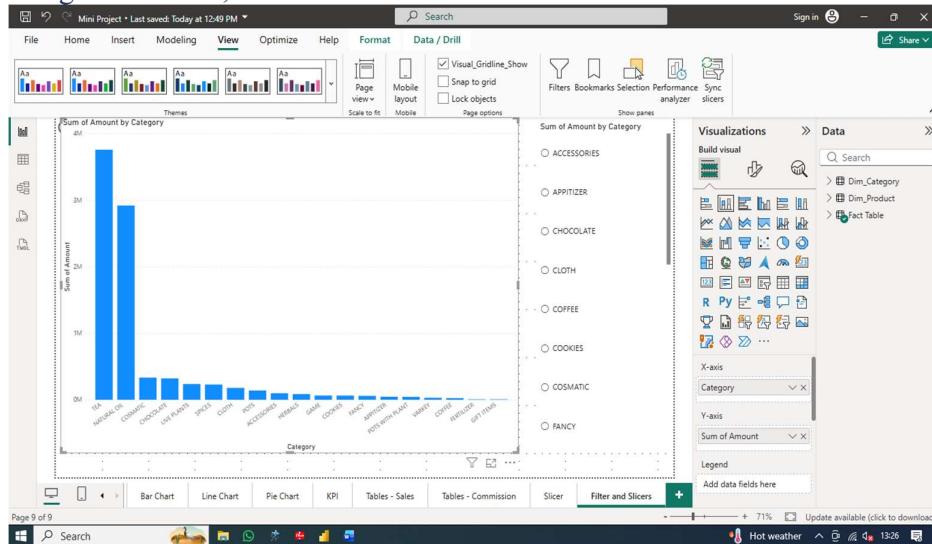
Total Sales by Category



Total Sales by Commission and Average Commission



using Drill-down, Filters and slicers



DASHBOARD



Descriptive analysis and insights

Overview of Dataset

The dataset contains sales transaction details including:

Date & Month

Product details (Name, Category, Code)

Quantity sold

Pricing (MRP, Gross, Discount, Final Amount)

Commission

Total Records: 32,078 rows

Key Metrics Summary

Total Sales Amount: ₹86,32,424

Total Gross Revenue: ₹95,72,196

Total Discount Given: ₹9,39,926

Total Quantity Sold: 3,50,368 units

Total Commission: ₹19,32,013

Insight:

Around 9.8% revenue is given as discount

Commission is a significant cost (~22% of sales)

Category-wise Performance

Top categories based on sales:

Category	Sales Amount	Quantity
TEA	₹37,59,202	17,653
NATURAL OIL	₹29,20,275	11,705
COSMATIC	₹3,32,928	1,538
CHOCOLATE	₹3,20,828	2,22,867
LIVE PLANTS	₹2,36,443	1,740

Insights:

TEA is the highest revenue-generating category

CHOCOLATE has very high quantity but low revenue → low price items

NATURAL OIL also performs strongly

Monthly Sales Analysis

Top performing months:

Month	Sales
February 2026	₹22,51,892
May 2025	₹10,65,520
April 2025	₹10,32,503
December 2025	₹7,82,855
June 2025	₹6,50,630

Insights:

February 2026 shows peak sales (possibly seasonal demand)

Sales are not consistent → fluctuations present

Some months contribute significantly more than others

Pricing & Discount Insights

Discounts applied across most transactions

Total discount ~ ₹9.39 Lakhs

Average discount per transaction is moderate

Insight:

Discounts are used as a sales-driving strategy

Need to analyze if discounts are impacting profit

Quantity Analysis

Total units sold: 3.5 lakh+

High volume sales in low-cost products (like chocolate)

Insight:

Business has a mix of:

High-volume low-value products

Low-volume high-value products

Commission Analysis

Total commission: ₹19,32,013

Insight:

Commission is a major expense

Could impact net profitability significantly

Overall Insights

- ✓ TEA and NATURAL OIL are key revenue drivers
- ✓ High discounts indicate aggressive marketing strategy
- ✓ Sales are seasonal and uneven
- ✓ Some products sell in bulk but generate low revenue
- ✓ Commission cost is relatively high

Diagnosis analysis and insights

Overall Business Performance

- Total Sales: ₹86,32,424
- Total Quantity Sold: 3,50,368 units
- Average Order Value: ₹269
- Total Discount Given: ₹9,39,926
- Total Commission Paid: ₹19,32,013

Insight:

- Discounts are ~10.8% of total sales, which is quite high.
- Commission is ~22% of sales, which impacts profitability significantly.

Diagnosis:

Profit margins may be under pressure due to high commission + discount combination.

Category-wise Performance

Top Categories by Sales:

1. TEA – ₹37.5 Lakhs
2. NATURAL OIL – ₹29.2 Lakhs
3. COSMETIC – ₹3.3 Lakhs
4. CHOCOLATE – ₹3.2 Lakhs
5. LIVE PLANTS – ₹2.3 Lakhs

Insights:

- TEA + NATURAL OIL = ~78% of total revenue
- CHOCOLATE has very high quantity but low revenue

Diagnosis:

- Business is heavily dependent on 2 categories (risk factor)
 - Chocolate is a low-value, high-volume product → may reduce profit margin
-

Monthly Sales Trend

Top Months:

- Feb 2026: ₹22.5 Lakhs (Highest)
- May 2025: ₹10.6 Lakhs
- April 2025: ₹10.3 Lakhs

Insights:

- Sudden spike in Feb 2026
- Some months show low or inconsistent performance

Diagnosis:

- Likely reasons:
 - Seasonal demand (festivals / offers)
 - Promotions or discounts
 - Bulk orders or campaigns
-

Discount Impact Analysis

- Total Discount: ₹9.39 Lakhs

Insights:

- Discounts are not evenly distributed
- High sales months likely have higher discounts

Diagnosis:

- Sales growth may be discount-driven, not demand-driven
 - Risk: Customers may become price-sensitive
-

Commission Analysis

- Total Commission: ₹19.3 Lakhs

Insights:

- Commission is very high compared to sales
- Some products may have higher commission rates

Diagnosis:

- Profit leakage through:
 - Sales channels
 - Agents / partners
-

Product-Level Issues

Observations:

- Some products:
 - High quantity but low revenue
 - High discount but low return

Diagnosis:

- Possible issues:
 - Poor pricing strategy
 - Overstocked or slow-moving items
 - Inefficient product mix
-

Key Problems Identified

1. Over-dependence on TEA & NATURAL OIL
 2. High discounts reducing profitability
 3. High commission expenses
 4. Uneven monthly sales (unstable growth)
 5. Low-value high-volume products (like Chocolate)
-

Predictive analysis and insights

Key Business Insights

- Strong Dependency
- Amount is directly derived from Gross and Discount

Discount Impact

- Every ₹1 increase in discount reduces revenue by almost ₹1
High discounts = direct loss in revenue

Commission Influence

- Commission has **negligible effect** on final Amount
It does NOT significantly impact revenue

Reduce Discount Dependency

- Try controlled discount strategies
Improves profit directly

Focus on Increasing Gross Sales

- More sales volume = more revenue
Without increasing discounts

Prescriptive analysis and insights

Category Performance Insight

- **Top performing category:** TEA (₹37.5L+ revenue)
- **Second:** NATURAL OIL (₹29L+)
- Low performers: Fertilizer, Gift Items, Coffee

Prescriptive Actions

- Increase stock and promotions for **TEA & NATURAL OIL**
 - Reduce inventory or redesign strategy for **low-performing categories**
 - Bundle low-performing items with best sellers (e.g., Tea + Cookies combo)
-

Product-Level Insight

- **Top product:** Rosemary Essential Oil 30ML (₹20L+ revenue)
- Other strong products:
 - White Tea 50G
 - Masala Chai 250G
 - Attar Premium
 - Eucalyptus Oil

Prescriptive Actions

- Focus marketing campaigns on **top 5 products**
 - Introduce **premium pricing strategy** for high-demand items
 - Ensure **zero stock-outs** for these products
 - Launch **similar variants** (e.g., new essential oils)
-

Monthly Sales Trend Insight

- Peak month: **February 2026 (₹22.5L+)**
- Strong months: May, April
- Weak months: September, November

Prescriptive Actions

- Identify why February performed well → replicate strategy
 - Offer **seasonal discounts** during low months
 - Plan **festive or promotional campaigns** for weak periods
-

Discount Impact Insight

- Discounts are consistently applied across transactions
- High discounts may reduce profit margins

Prescriptive Actions

- Apply **targeted discounts only on slow-moving products**
 - Avoid giving discounts on **high-demand items**
 - Use **dynamic pricing strategy**
-

Commission Insight

- Commission varies per product
- High commission reduces net profitability

Prescriptive Actions

- Optimize commission structure:
 - Lower commission on high-selling products
 - Incentivize sales team to push low-performing items
-

Quantity vs Revenue Insight

- Some products generate high revenue with low quantity → high-value items
- Others need high volume → low-margin products

Prescriptive Actions

- Focus on **high-margin products** for profit growth
 - Promote **bulk sales offers** for low-margin items
-

Category Diversification Insight

- Revenue is heavily dependent on few categories (Tea & Oil)

Prescriptive Actions

- Expand other categories like:
 - Cosmetics
 - Chocolates
 - Herbal products
 - Reduce dependency risk by diversifying revenue sources
-

Final Business Recommendations

Growth Strategy

- Double down on **Tea & Essential Oils**
- Expand **premium product line**

Profit Optimization

- Reduce unnecessary discounts
- Control commission expenses

Sales Strategy

- Use **combo offers & cross-selling**
- Seasonal campaigns for weak months

Inventory Strategy

- Maintain high stock for fast-moving items
 - Reduce dead stock
-

Conclusion

The business should **focus on high-performing categories (Tea & Oils), optimize pricing and discounts, and improve sales strategies for low-performing products.**

By applying targeted promotions, better inventory planning, and commission control, overall **revenue and profitability can be significantly improved.**
